

SSB utility

Model (conjectured) — human; produced by ZG.

0.1 Package

- **Completeness.** For all $p, q \in \Delta(X)$: $p \succcurlyeq q$ or $q \succcurlyeq p$.
- **Closed-graph continuity.** The set $\{(p, q) \in \Delta(X) \times \Delta(X) : p \succcurlyeq q\}$ is closed in the product topology.
- **Archimedean.** For all $p, q, r \in \Delta(X)$: if $p \succ q \succ r$ then there are $\alpha, \beta \in (0,1)$ with $\alpha p + (1-\alpha)r \succ q$ and $q \succ \beta p + (1-\beta)r$.
- **$\neg$ Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.
- **$\neg$ Independence.** For all $p, q, r \in \Delta(X)$ and $\lambda \in (0,1]$: $p \succcurlyeq q$ if and only if $\lambda p + (1-\lambda)r \succcurlyeq \lambda q + (1-\lambda)r$.

0.2 Notes

Sample conjecture; which of betweenness and quasi-transitivity hold is to be checked.

0.3 Sources

- Fishburn (1982), Nontransitive measurable utility, Journal of Mathematical Psychology 26

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